

AI Super-Sleuth Finds 100+ Hidden Planets, Including a World That Orbits in Just One Day!

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Have you ever tried to find a tiny firefly next to a giant spotlight? That's what **planet hunters** do! They look for tiny **dips** in starlight when a planet passes in front of its star. Now, a clever AI named **RAVEN** has found over 100 hidden planets in data from NASA's **TESS** telescope. **TESS** watched more than 2.2 million stars for four years. **RAVEN**, built by scientists at the University of Warwick, is like a super-smart detective. It can tell if a **dip** in light is really a planet or just a trick, like two stars crossing. The AI found 118 new planets and over 2,000 possible ones. Some are really extreme! There are planets that zoom around their star in less than a day. Others are in the "**Neptunian desert**"a place where scientists thought planets couldn't exist. The team also found systems with multiple planets packed close together. Dr. Marina Lafarga Magro, who led the study, said **RAVEN** helps find the best planets to study later. The AI was trained on hundreds of thousands of fake planets and fake signals. This helps it spot real planets with amazing accuracy. The scientists also learned how common **close-in planets** are. About 9-10% of Sun-like stars have a planet orbiting very close. That's a much more precise number than before. Dr. David Armstrong said **RAVEN** doesn't just list planetsit gives a reliable map of what kinds of planets are out there. The team shared their tools so other scientists can explore the data too. Who knows what other weird worlds are waiting to be found?

Word Treasure

TESS -- A NASA space telescope that watches millions of stars for tiny dips in light that could be planets.

RAVEN -- A smart computer program trained to spot real planets among thousands of confusing signals.

Neptunian desert -- A region very close to a star where scientists once thought Neptune-sized planets could never exist.

planet hunters -- Scientists who search for planets around other stars by watching for small changes in starlight.

dip -- A small, short drop in a star's brightness that happens when a planet passes in front of it.

close-in planets -- Planets that orbit so tightly around their star they can complete a full year in just days.

Background you need

NASA's TESS (Transiting Exoplanet Survey Satellite) was launched in 2018 and has discovered thousands of planet candidates by watching the brightness of over 200,000 nearby stars.

The first exoplanet around a Sun-like star was found in 1995, and since then astronomers have confirmed over 5,500 exoplanets.

The 'Neptunian desert' is a real puzzle--a shortage of Neptune-sized worlds in orbits very close to their stars that scientists are still trying to explain.

Break it down

WHO: Dr. Marina Lafarga Magro and her team at the University of Warwick.

WHAT: The AI RAVEN analyzed TESS data and found 118 new planets plus over 2,000 possible ones.

WHERE: In space, using data from NASA's TESS telescope, which watches the entire sky.

WHY: To build a reliable map of what kinds of planets exist and find the best ones for further study.

Why it matters

Discovering new planets helps us learn if Earth-like worlds are common, which could inspire your future studies in science or even a career searching for life beyond our solar system.

Test what you remember

1. What is RAVEN?

- ☐ (A) a new planet
- ☐ (B) a NASA telescope
- ☐ (C) an AI that finds planets
- ☐ (D) a star system

2. How many new planets did RAVEN discover in the TESS data?

- ☐ (A) 10
- ☐ (B) 50
- ☐ (C) 118
- ☐ (D) 2,000

3. Which NASA space telescope provided the data RAVEN used?

- ☐ (A) Hubble
- ☐ (B) TESS
- ☐ (C) James Webb
- ☐ (D) RAVEN

4. What is the 'Neptunian desert'?

- ☐ (A) a real desert on the planet Neptune
- ☐ (B) a place where scientists thought Neptune-sized planets couldn't exist
- ☐ (C) a dry planet found by RAVEN
- ☐ (D) a type of telescope used in the study

5. Who led the research that built RAVEN?

- ☐ (A) Dr. Shirley Xerxes
- ☐ (B) Dr. Marina Lafarga Magro
- ☐ (C) Dr. David Armstrong
- ☐ (D) Dr. Nicola Bortolotti

6. According to the study, about how many Sun-like stars have a close-in planet?

- ☐ (A) 1%
- ☐ (B) 9-10%
- ☐ (C) 25%
- ☐ (D) 50%

Answer key (try the quiz first!): 1: C 2: C 3: B 4: B 5: B 6: B

Questions to think about

1. Planet hunter: RAVEN acts like a super sleuth, saving us years of work by spotting real planets among thousands of fake or confusing signals.
2. AI researcher: Training RAVEN on hundreds of thousands of fake planets and signals made it extremely accurate, showing how AI can solve complex pattern problems in science.
3. Astronomer: Knowing that 9-10% of Sun-like stars have a close-in planet gives us a much more precise and reliable picture of our galaxy.

MY THOUGHTS (at least 20 words):